

HDOC 35

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Batch: 16

1. Algorithm

1. Start main program.
2. Read dimension n for a square matrix $mat[n][n]$ and input its elements.
3. Display the menu with options 1 to 4 and read choice.
4. Case 1 (Distinct Diagonal Elements): Loop through diagonal elements $mat[i][i]$. Compare each pair. If a match is found, print "not distinct", otherwise "all distinct".
5. Case 2 (Sum of Diagonal Elements): Loop ifrom 0 to $n-1$ and add $mat[i][i]$ to a running sum. Print the sum.
6. Case 3 (Zigzag Diagonal Traversal): Loop dfrom 0 to $(2*n)-2$. If dis even, set starting indices for upward-right traversal and iterate until bounds are reached. If dis odd, set starting indices for downward-left traversal and iterate until bounds are reached. Print each element visited.
7. Case 4 (Exit): Terminate the program loop.
8. Stop

2. Test-Case Table

TC ID	Scenario	Inputs	Expected Output
TC-01	Distinct check	$n=3$, $Mat=[1\ 2\ 3; 4\ 5\ 6; 7\ 8\ 9]$, choice: 1	All the main diagonal elements are distinct.
TC-02	Sum of diagonal	choice: 2	Sum of main diagonal elements: 15
TC-03	Zigzag Traversal	choice: 3	Diagonal traversal: 1 2 4 7 5 3 6 8 9
TC-04	Not Distinct	$n=2$, $Mat=[3\ 4; 5\ 3]$, choice: 1	Main diagonal elements are not distinct.
TC-05	Exit	Choice: 4	Exiting program...

3. C Program Source Code

```
#include <stdio.h>

void checkDistinctDiagonal(int n, int mat[n][n]); void sumOfDiagonal(int n,
int mat[n][n]);
void diagonalTraversal(int n, int mat[n][n]);

int main() {
    int n;
    printf("Enter the size of the square matrix (n x n): "); scanf("%d", &n);
    if (n <= 0) return 1;

    int mat[n][n];
    printf("Enter the matrix elements:\n"); for (int i = 0; i < n;
i++) {
        for (int j = 0; j < n; j++) { scanf("%d", &mat[i][j]);
        }
    }

    int choice; do {
        printf("\n--- Square Matrix Operations Menu ---\n"); printf("1. Check if main diagonal
elements are distinct\n"); printf("2. Find the sum of main diagonal elements\n"); printf("3.
Display diagonal traversal\n");
        printf("4. Exit\n"); printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) { case 1:
            checkDistinctDiagonal(n, mat); break;
            case 2:
            sumOfDiagonal(n, mat); break;
            case 3:
            diagonalTraversal(n, mat); break;
            case 4:
            printf("Exiting program...\n"); break;
            default:
            printf("Error: Invalid choice.\n");
        }
    } while (choice != 4);

    return 0;
}

void checkDistinctDiagonal(int n, int mat[n][n]) { int distinct = 1;
    for (int i = 0; i < n; i++) {
```

```

        for (int j = i + 1; j < n; j++) { if (mat[i][i] ==
            mat[j][j]) {
                distinct=0; break;
            }
        }
        if (!distinct) break;
    }

    if (distinct) {
        printf("All the main diagonal elements are distinct.\n");
    } else {
        printf("Main diagonal elements are not distinct.\n");
    }
}

void sumOfDiagonal(int n, int mat[n][n]) { int sum = 0;
    for (int i = 0; i < n; i++) { sum += mat[i][i];
    }
    printf("Sum of main diagonal elements: %d\n", sum);
}

void diagonalTraversal(int n, int mat[n][n]) { printf("Diagonal traversal: ");
    for (int d = 0; d < 2 * n - 1; d++) { if (d % 2 == 0) {
        int r = (d < n) ? d : n - 1;
        int c = (d < n) ? 0 : d - n + 1; while (r >= 0 && c < n) {
            printf("%d ", mat[r][c]); r--;
            c++;
        }
    } else {
        int r = (d < n) ? 0 : d - n + 1;
        int c = (d < n) ? d : n - 1; while (r < n && c >=
        0) {
            printf("%d ", mat[r][c]); r++;
            c--;
        }
    }
    }
    printf("\n");
}

```

4. Compilation and Execution

```
(kali㉿kali)-[~/liak]
$ ./a.out
Enter the size of the square matrix (n x n): 3 3
Enter the matrix elements:
1 2 3
4 5 6
7 8 9

— Square Matrix Operations Menu —
1. Check if main diagonal elements are distinct
2. Find the sum of main diagonal elements
3. Display diagonal traversal
4. Exit
Enter your choice: Error: Invalid choice.

— Square Matrix Operations Menu —
1. Check if main diagonal elements are distinct
2. Find the sum of main diagonal elements
3. Display diagonal traversal
4. Exit
Enter your choice: 2
Sum of main diagonal elements: 15

— Square Matrix Operations Menu —
1. Check if main diagonal elements are distinct
2. Find the sum of main diagonal elements
3. Display diagonal traversal
4. Exit
Enter your choice: 1
All the main diagonal elements are distinct.

— Square Matrix Operations Menu —
1. Check if main diagonal elements are distinct
2. Find the sum of main diagonal elements
3. Display diagonal traversal
```